

# Autodesk® **Scaleform**®

## Scaleform

Scaleform 4.3   Scaleform

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3.01  
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### Autodesk® Scaleform® 4.3

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## Autodesk Scaleform

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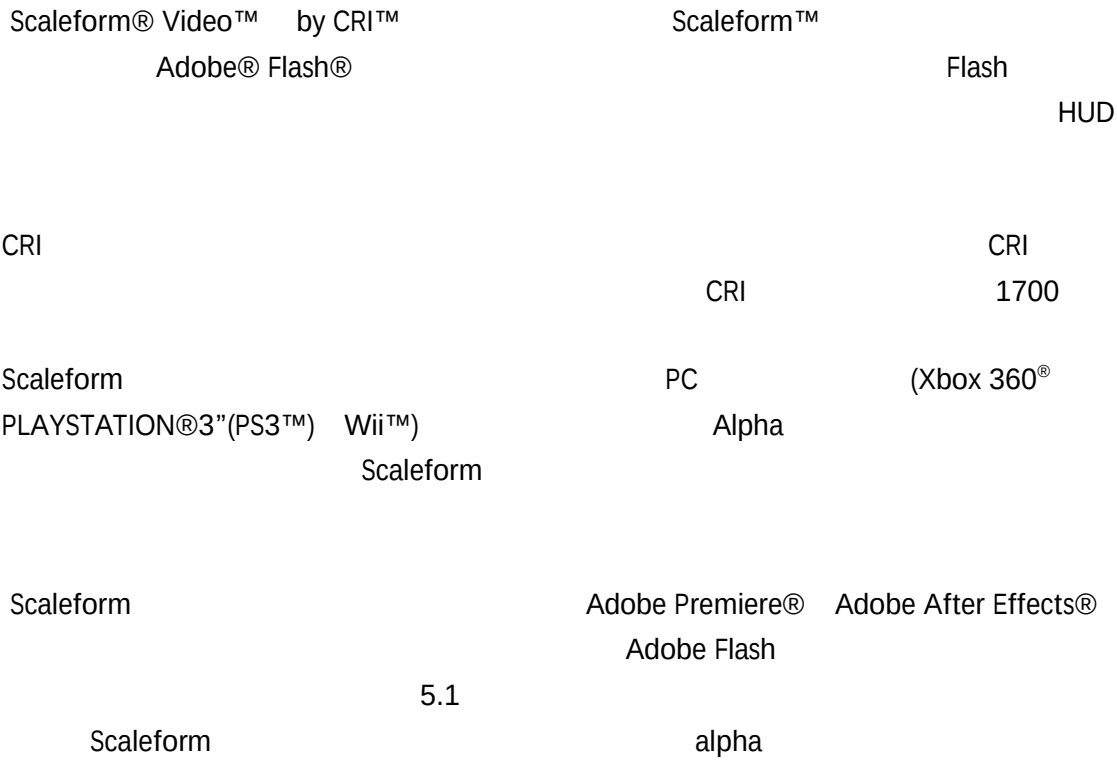
(301) 446-3200

(301) 446-3199

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1



Scaleform Video SDK

- Scaleform
- Scaleform
- /
- Scaleform ( )

## 2      **Scaleform**

Scaleform

### **2.1**

Scaleform

*C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder*

Scaleform

*C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video*

Flash  
*videodemo.swf*

USM

Scaleform

## 2.2

- *getting\_started\_with\_video\_tutorial.flv* – Flash
- *getting\_started\_with\_video\_tutorial.swf* – Flash
- *sample.avi* –
- *sample\_audio.wav* –
- *sample\_cue\_points.txt* –
- *sample\_subtitles.txt* –

## 2.3

### Scaleform

- *Medianoche.exe* —
- *ScaleformVideoEncoder.exe* — ( )
- *TMPGLib.dll* —
- *VideoEncoderUtil.dll* — *ScaleformVideoEncoder.exe*
- *VideoPlayer.swf* — Scaleform *Preview*



## 2.4

- 

1080p API (HD)

- 

- 

SDK

## 2.5

Scaleform®  
Scaleform

CRI Movie™

Adobe Flash®

3D

HUD

Scaleform  
Adobe Flash

Scaleform  
CRI Movie™

Scaleform™ 4.1

## 2.6

- : AVI WAV
- : Scaleform Xbox 360 PS3 PC Wii
- :
- :  
Scaleform
- **Alpha** : alpha  
“ ”
- : 3D
- : 32
- : using Scaleform Flash 4.1 ActionScript™
- : Scaleform ActionScript  
UI
- : / 5.1
- :
- : Scaleform Video
- :  
Scaleform USM  
Flash ActionScript

## 2.7

Scaleform

Scaleform CRI

USM

USM

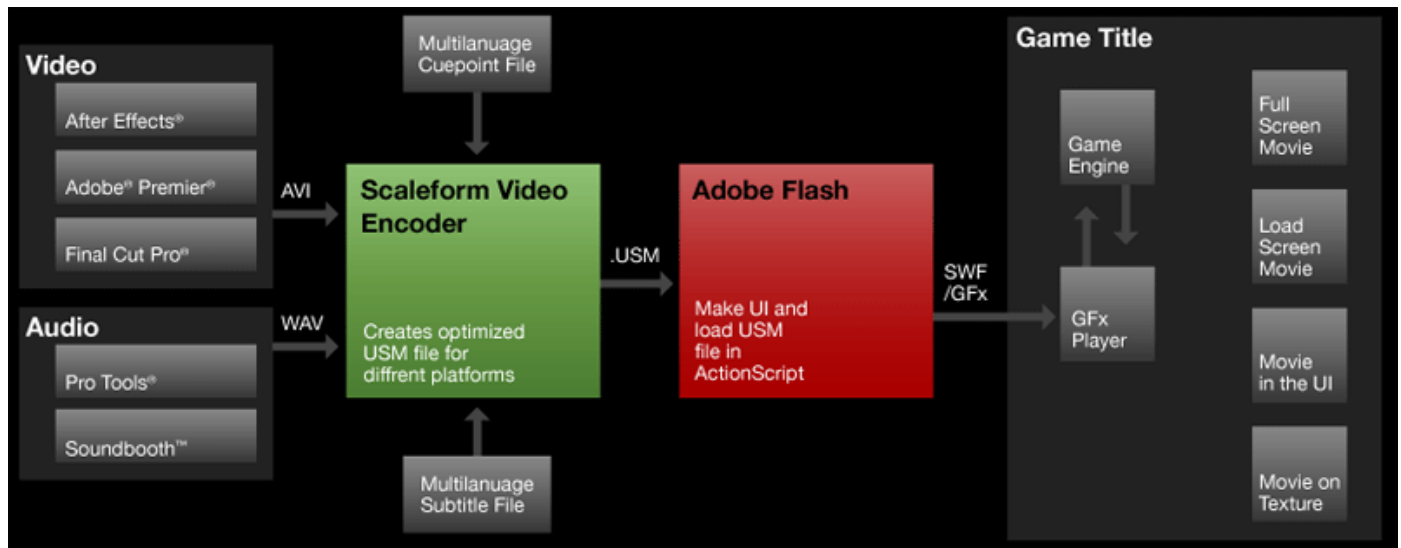
Scaleform

Scaleform

USM

*C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video\scaleform\_logo.usm*

## 2.8



### 1 Scaleform

1. Scaleform API.
2. Adobe Premiere AVI
3. Scaleform AVI USM
4. Adobe Flash SWF USM
5. ActionScript
6. SWF
7. SWF

\* USM Flash ActionScript

Flash

### 3 Scaleform

Scaleform™ 4.3

CRI

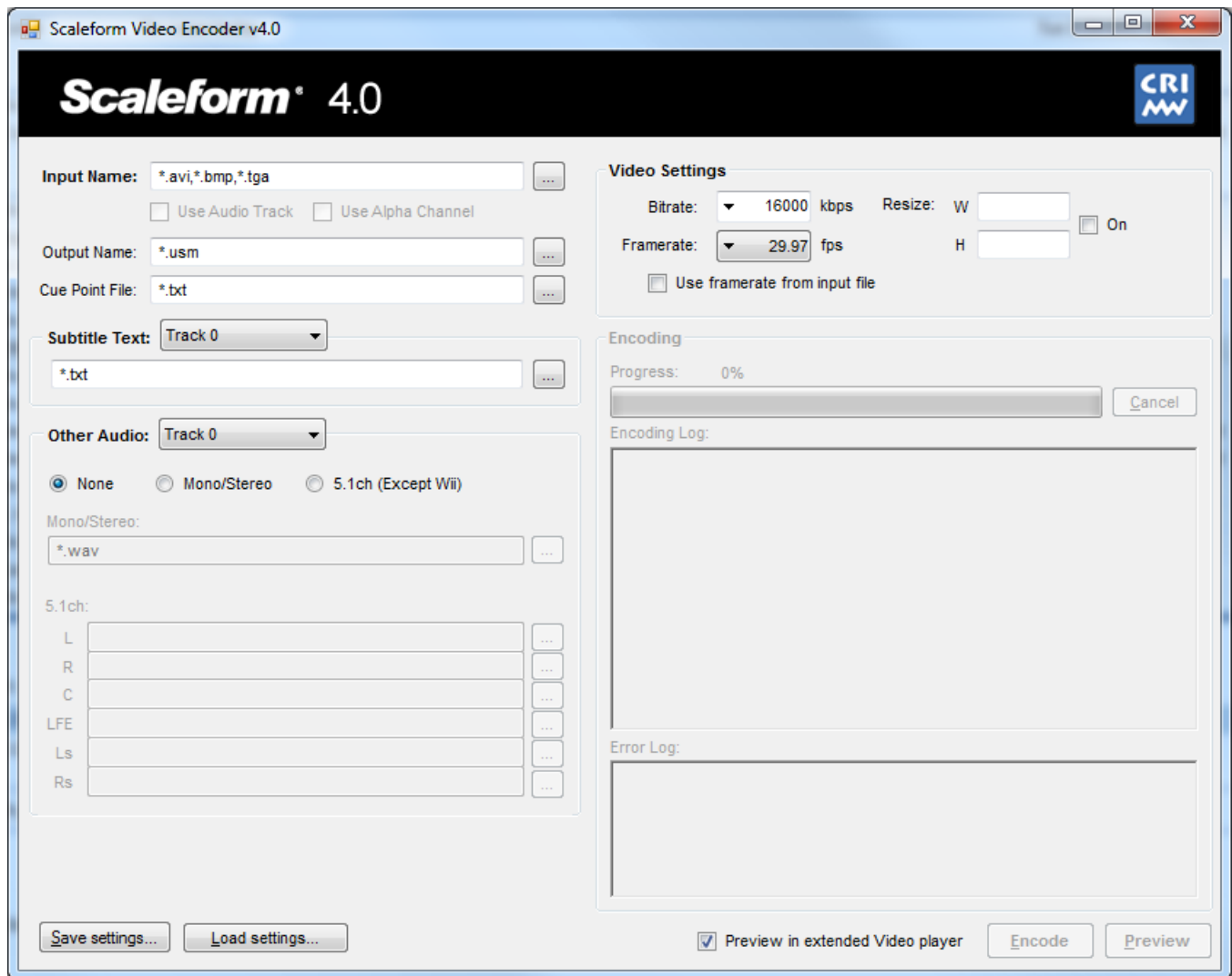
USM

CRI

Scaleform

Scaleform

USM



2: Scaleform

### 3.1

USM

USM

alpha

2

Windows™

## Scaleform

Scaleform

**Windows Vista:** *Start->All Programs->Scaleform->GFX SDK 4.3->Video*

**Windows XP:** *Start->Programs->Scaleform->GFX SDK 4.3->Video*

1. *Browse*  AVI
  - a. AVI BMP TGA
  - b. BMP TGA

**Input Name:**

☐ Use Audio Track    ☐ Use Alpha Channel

**Output Name:**

**3:**

- 
- ```

graph TD
    S1[1. Select Input File] --> S2[2. Select Audio Format]
    S2 --> S3[3. Select Alpha Channel]
    S3 --> S4[4. Select Output Name]
    S4 --> S5[5. Encode Audio Track]
  
```
2. **Select Audio Format**  
 BMP TGA AVI  
*Use Audio Track*
3. **Select Alpha Channel**  
 alpha USM  
*Use Alpha Channel*
4. **Select Output Name**  
 USM browse  
*Output Name*
5. **Encode Audio Track**  
*' Encode '*

## 3.2

- 1.
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\Tools\VideoEncoder\sample.avi*
3. *Encode to*
4. *Preview* *USM*

### 3.3

SWF

USM

Cue Point File

Browse

Cue Point File:

4:

10000

5000

5

—  
— 5000/1000=5

DVD

=

Flash

[ActionScript](#)

time, cue point type (0 or 1), cue point name, parameter1, parameter2, ...  
paramter10

:

1000, 0, cue\_point\_1, my\_param=5, my\_other\_param=10

#### CuePoint.txt

```
1000
0,0,start
1000,0,cue1
2000,1,cue2,name1_0=value1_0,name1_1=value1_1
3000,0,cue3
4000,1,cue4,name4_0=value4_0
5000,0,cue5
6000,1,cue6,name6_0=value6_0,name6_1=value6_1
```



```

7000,0,cue7,name7_0=value7_0,name7_1=value7_1,name7_2=value7_2
8000,1,cue8,name8_0=value_0
9000,0,cue9
10000,1,cue10,name10_0=value10_0,name10_1=value10_1
11000,0,end

```

### 3.3.1

1.       Scaleform
2.       *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3.       *Cue Point File*               *browse*               *sample\_cue\_points.txt*       —  
          *sample.avi*
4.       *Encode*
5.       *Preview*                       USM
6.       *fast forward*   *rewind*



### 3.4.1

1. Scaleform
2. *C:\Program Files\Scaleform\GFx SDK 4.3 \Bin\Tools\VideoEncoder\sample.avi*
3. Cue Point File *sample.avi* browse *sample\_cue\_points.txt* —
4. Subtitle Text *sample.avi* — browse *sample\_subtitles.txt*
5. Encode
6. , Scaleform Player (Scaleform ),  
*getting\_started\_with\_video\_tutorial.swf*
7. 2-5 8-11

3.5

/ 5.1

1. Scaleform Other Audio
2. — / 5.1 Wii  
5.1 Wii 5.1
3. / Mono/Stereo Browse WAV
4. Use Audio Track

Other Audio: Track 0

☒ None

☐ Mono/Stereo

☐ 5.1ch (Except Wii)

Mono/Stereo:

\*.wav

...

5.1ch:

L

R

C

LFE

Ls

Rs

...

...

...

...

...

...

6:

5. 5.1 Browse — L R C LFE Ls Rs —

5.1

L —

R —

C — ( )

LFE —

Ls –  
Rs –

### 3.5.1

## 3.6

*Bitrate*

**Video Settings**

Bitrate:  kbps    Resize: W  ☐ On

Framerate:  fps    H

☐ Use framerate from input file

**7:**

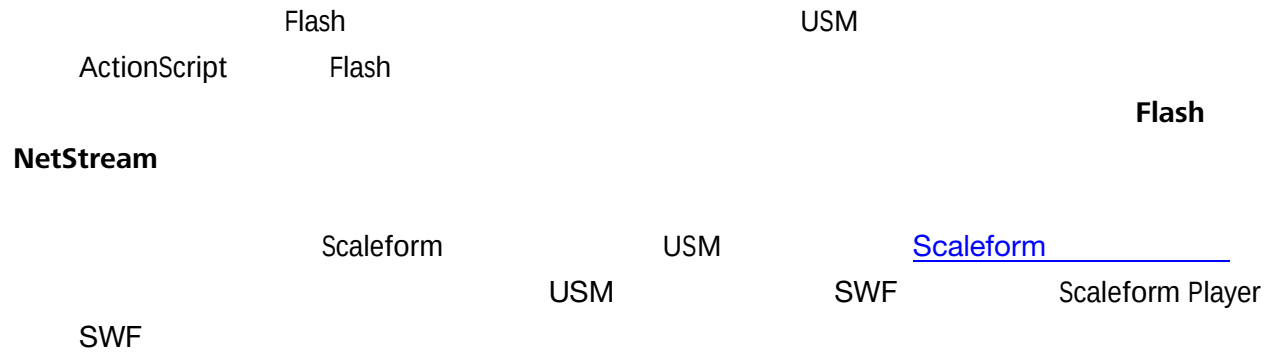
1. *Bitrate*    —

| (kbps) |       |         |      |       |
|--------|-------|---------|------|-------|
|        | PS3   | Xbox360 | Wii  | PC    |
| 1080p  | 40000 | 40000   | n/a  | 40000 |
| 1080p  | 30000 | 30000   | n/a  | 30000 |
| 1080p  | 20000 | 20000   | n/a  | 20000 |
| 720p   | 36000 | 36000   | n/a  | 36000 |
| 720p   | 26000 | 26000   | n/a  | 26000 |
| 720p   | 16000 | 16000   | n/a  | 16000 |
| 480p   | 16000 | 16000   | 8000 | 16000 |
| 480p   | 12000 | 12000   | 6000 | 12000 |
| 480p   | 8000  | 8000    | 4000 | 8000  |

2.                      *Framerate*                                              29.97 fps—                      NTSC                      AVI  
                             *Use framerate from input file*



## 4 Flash





## 4.1 Flash

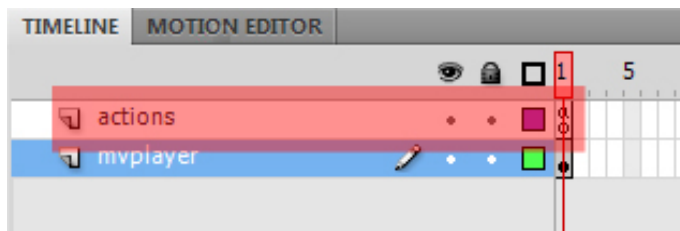
AS 2.0 Flash  
SWF

Scaleform

USM

1. *Layer 1* *mvplayer*

2. *New Layer* *' action '*



8: *mvplayer*

3. *actions* F9 *Actions* *Actions*

```
_global.gfxExtensions = true;
_focusrect = false;
onLoad = function()
{
    mvplayer.playVideo("sample.usm");
}
```

SWF

USM

*" C:/pathtovideo/sample.usm "*

4. *mvplayer*

5.

a.

b.

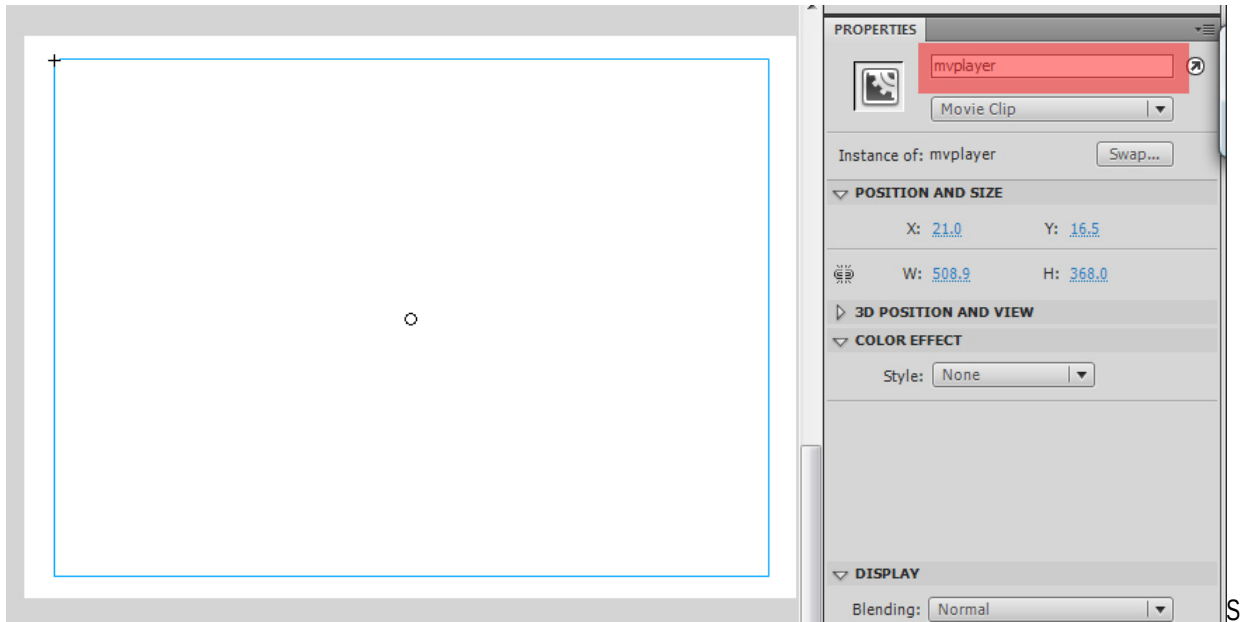
c.

*Convert to Symbol*

d.

*' mvplayer '*

OK



9: **mvplayer**

6. `new mvplayer`

7. Properties `' mvplayer '`

8. `mvplayer`

9. `mvplayer` `' actions '`

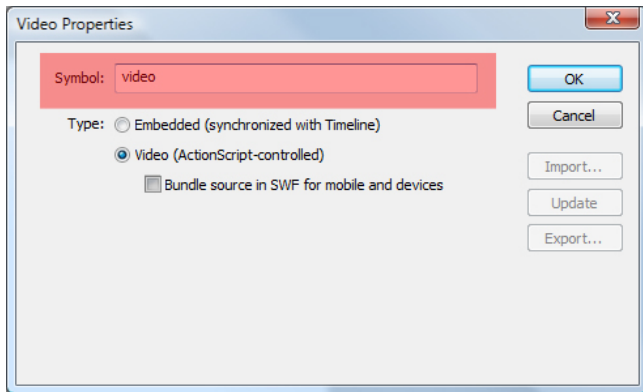
10. `actions` 1 `Actions`

```
nc = new NetConnection();
nc.connect(null);
ns = new NetStream(nc);
video.attachVideo(ns);
this.attachAudio(ns);
var audio_sound:Sound = new Sound(this);
function playVideo(videoToPlay:String):Void
{
    ns.play(videoToPlay);
}
```

`mvplayer` `actions` 1 `Layer 1`

11. `Library` `New Video`

12. `' Video '` `OK`



10:

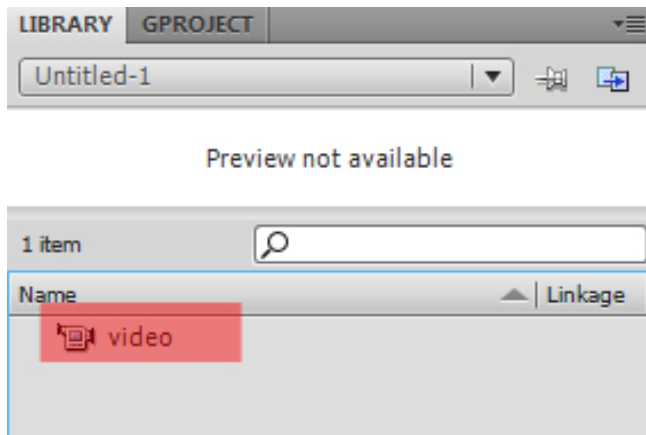
13. *New Layer* button

‘ Layer 3 ’

14. *Library*

*Layer 3*

*Library*



11: **Library**

15. *Properties*

‘ Video ’

16.

## 4.2

- |    |       |           |                  |
|----|-------|-----------|------------------|
|    |       | SWF       | Scaleform Player |
| 1. | Flash | Scaleform | SWF              |
- 
- |    |  |                  |
|----|--|------------------|
| 1. |  | Scaleform Player |
|----|--|------------------|
- 
- |    |     |                      |
|----|-----|----------------------|
| 2. | SWF | Scaleform Player     |
|    | a.  | SWF Scaleform Player |

## 5 ActionScript

Flash

USM

ActionScript

Flash

Flash SWF

Scaleform

ActionScript

## 5.1 ActionScript

USM

/

Flash

Flash SWF

/

USM

USM

Flash

ActionScript

*mvplayer*

*actions* 1

1. *mvplayer*

2. *mvplayer* *actions* *Actions*

3. *traceCuePoint*

*ns.onCuePoint* = *traceCuePoint*;

4. *traceCuePoint* *metaPropPJParams*

(*currentCuePoint.name*) (*currentCuePoint.time*)  
(*currentCuePoint.type*) *ns.time*

```
function traceCuePoint(currentCuePoint:Object):Void
{
    var metaPropPJParams:Object = currentCuePoint.parameters;
    trace("\t\t name: " + currentCuePoint.name);
    trace("\t\t time: " + currentCuePoint.time);
    trace("\t\t type: " + currentCuePoint.type);
    if (metaPropPJParams != undefined)
    {
        trace("\t\t parameters:");
        traceObject(metaPropPJParams, 4);
    }
}
```

5. *traceObject*

SWF

```

function traceObject(obj:Object, indent:Number):Void
{
    var indentString:String = "";
    for (var j:Number = 0; j < indent; j++)
    {
        indentString += "\t";
    }
    for (var i:String in obj)
    {
        if (typeof(obj[i]) == "object")
        {
            trace(indentString + " " + i + ": [Object]");
            traceObject(obj[i], indent + 1);
        }
        else
        {
            trace(indentString + " " + i + ": " + obj[i]);
        }
    }
}

```

|    |                  |                  |           |     |
|----|------------------|------------------|-----------|-----|
| 6. | Flash            | Scaleform Player | Scaleform | SWF |
|    | Scaleform Player |                  |           |     |
|    | Scaleform Player |                  |           |     |

## 5.2 ActionScript

|                   |                  |                                                                                                                                                                                                                                                                                               |              |              |                |  |                     |       |  |
|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|----------------|--|---------------------|-------|--|
|                   | Flash            |                                                                                                                                                                                                                                                                                               |              |              |                |  |                     |       |  |
|                   |                  | Flash SWF                                                                                                                                                                                                                                                                                     | /            |              |                |  | USM                 |       |  |
| <i>sample.usm</i> |                  |                                                                                                                                                                                                                                                                                               | <u>Flash</u> |              | <u>Flash</u>   |  |                     |       |  |
| 1.                | <i>mvplayer</i>  |                                                                                                                                                                                                                                                                                               |              |              |                |  |                     |       |  |
| 2.                | <i>mvplayer</i>  | <i>actions</i>                                                                                                                                                                                                                                                                                |              |              | <i>Actions</i> |  |                     |       |  |
| 3.                |                  | subtitleChannelNumber                                                                                                                                                                                                                                                                         |              |              |                |  | 0                   |       |  |
|                   |                  | <code>var subtitleChannelNumber = 0;</code>                                                                                                                                                                                                                                                   |              |              |                |  |                     |       |  |
| 4.                |                  |                                                                                                                                                                                                                                                                                               |              |              |                |  | 0                   |       |  |
|                   |                  | <code>ns.onMetaData = function(infoObject:Object)</code><br><code>{</code><br><code>    ns.subtitleTrack = 1;</code><br><code>    subtitleChannelNumber = infoObject.subtitleTracksNumber;</code><br><code>    trace("Subtitle Channels: " + subtitleChannelNumber);</code><br><code>}</code> |              |              |                |  |                     |       |  |
| 5.                |                  | <i>mvplayer</i>                                                                                                                                                                                                                                                                               |              |              |                |  | <i>' subtitle '</i> |       |  |
| 6.                |                  |                                                                                                                                                                                                                                                                                               |              |              |                |  | USM                 | Msg   |  |
|                   |                  | subtitleId2                                                                                                                                                                                                                                                                                   |              | ActionScript |                |  |                     | Msg   |  |
|                   | 5                | <i>subtitle dynamic</i>                                                                                                                                                                                                                                                                       |              |              |                |  |                     |       |  |
|                   |                  | <code>ns.onSubtitle = function(msg:String)</code><br><code>{</code><br><code>    trace("Subtitle: " + msg);</code><br><code>    subtitle.text = msg;</code><br><code>}</code>                                                                                                                 |              |              |                |  |                     |       |  |
| 7.                | Flash            | Scaleform Player                                                                                                                                                                                                                                                                              |              | Scaleform    |                |  | SWF                 |       |  |
|                   | Scaleform Player |                                                                                                                                                                                                                                                                                               |              |              |                |  |                     | Flash |  |



## 5.3 ActionScript

USM

SWF

Flash SWF

/

USM

Flash

*mvplayer*

*actions*

1

*Actions*

1. *mvplayer*

2. *mvplayer* *actions*

*Actions*

3. AudioTracks

CurAudioTrack

0

```
var AudioTracks:Array;
var CurAudioTrack = 0;
```

4. audioTracks

audioTracks

```
        ' ns.onMetaData'
'AudioTracks = infoObject.audioTracks;' —
Channels: " + subtitleChannelNumber);'
ns.onMetaData = function(infoObject:Object)
{
    AudioTracks = infoObject.audioTracks;
}
        'trace("Subtitle
        '}'
```

5. *mvplayer*

‘prev\_audiotrack ’

6. ‘ mvplayer ’ ‘ action ’  
next\_audiotrack

*Actions*

```
prev_audiotrack.onRelease = function()
{
```

```

        if (AudioTracks == undefined || AudioTracks.length < 2)
        {
            return;
        }
        if (CurAudioTrack == 0)
        {
            CurAudioTrack = AudioTracks.length - 1;
        }
        else
        {
            CurAudioTrack--;
        }
        ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
    }

```

7.     *mvplayer*                             5                             ‘next\_audiotrack’

8.     ‘mvplayer’     ‘action’                             *Actions*  
        next\_audiotrack

```

next_audiotrack.onRelease = function()
{
    if (AudioTracks == undefined || AudioTracks.length < 2)
    {
        return;
    }
    CurAudioTrack++;
    if (CurAudioTrack >= AudioTracks.length)
    {
        CurAudioTrack = 0;
    }
    ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
}

```

9.     Scaleform Player                     Flash  
                                           *Next   Previous*                     USM

## 6 ActionScript

Scaleform    Flash            ActionScript  
                                         ‘ NetStream ’    Adobe

Scaleform  
NetStream

### 6.1                    *NetStream*

currentFps:Number

```
: var currentFps:Number = ns.currentFps;
```

time:Number

.

```
var currentTime:Number = ns.time;
```

### 6.2                    *NetStream*

```
onCuePoint = function(infoObject:Object) {}
```

USM

Adobe Flash Help            ‘ onCuePoint ’

```
onMetaData = function(infoObject:Object) {}
```

USM

Adobe Flash Help            ‘ onMetaData ’

#### **Scaleform Extensions**

audioTracks

USM

```

ns.onMetaData = function(infoObject:Object)
{
    audioTrackArray = infoObject.audioTracks;
}

```

subtitleTracksNumber

USM

```

ns.onMetaData = function(infoObject:Object)
{
    numSubtitleChannels = infoObject.subtitleTracksNumber;
    trace("Subtitle Channels: " + numSubtitleChannels);
}

```

```
onStatus = function(infoObject:Object) {}
```

NetStream

Adobe Flash Help      ‘ onStatus ’

## 6.3      ***NetStream***

close()

ns.time      0

```
ns.close();
```

pause([flag:Boolean])

```
ns.pause(true);
```

play(name:Object, start:Number, len:Number, reset:Object)

USM

```
ns.play("myVideo.usm");
```

`seek(offset: Number)`

`ns.seek(50);`

`setBufferTime(bufferTime: Number)`

`GFX::Video::Video`  
`(Bitrate)`

`, 1`  
`: ns.setBufferTime(2.5);`

## 6.4 *Scaleform*

`subtitleTrack = Number`

`NetStream`

`ns.subtitleTrack = 1;`

`audioTrack = Number`

`NetStream ( , )`  
`,`  
`: ns.audioTrack = 2;`

`subAudioTrack = Number`

`SubAudio , NetStream`

`: ns.subAudioTrack = 3;`

`voiceTrack = Number`

`5.1ch 5.1ch`

`: ns.voiceTrack = 1;`

`setReloadThresholdTime(reloadTime: Number)`

```

        4 ( setBufferTime), (Reload Threshold) 1 ,
Gfx::Video::Video 4 3 ,
        , Gfx::Video::Video

: ns.setReloadThresholdTime(1);

setNumberOfFramePools(numPools: Number)
        Gfx::Video::Video
        CPU 2
: ns.setNumberOfFramePools(3);

setOpenTimeout(timeout: Number)
        (USM ) 5 (
), 0
: ns.setOpenTimeout(5);

currentFrame: Number
        ( )。
: trace("Current FPS: " + ns.currentFps + " fps");

Loop = Boolean

        true NetStream false

        ns.loop = true;

clipRect = new rectangle(x: Number, y: Number, width: Number, height: Number)

        ns.clipRect = new rectangle(100, 100, 240, 200);

onSubtitle = function(msg: String) {}

        USM

        ns.onSubtitle = function(msg: String) { trace(msg); }

```

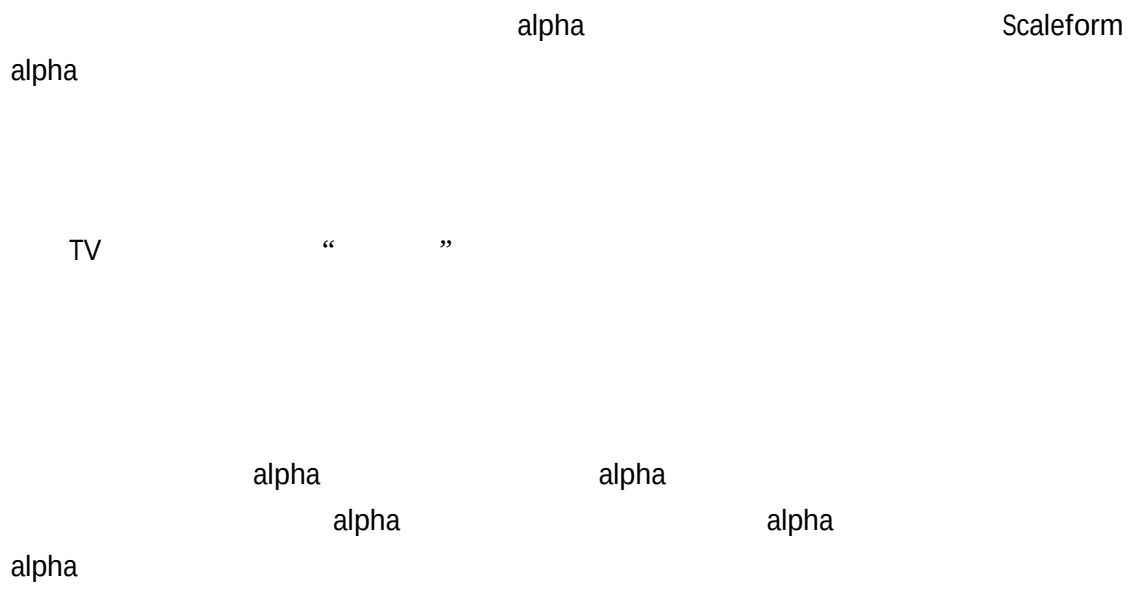
7



/



## 7.1 Alpha



# 8

Scaleform

## 8.1 Scaleform

|                                               |                            |                             |     |
|-----------------------------------------------|----------------------------|-----------------------------|-----|
| Scaleform                                     | <a href="#">Gfx::Audio</a> | <a href="#">Gfx::Loader</a> |     |
| <a href="#">SoundRenderer</a>                 | SWF                        | SoundRenderer               | C++ |
|                                               | Scaleform                  | FMOD                        |     |
| Sound::SoundRendererFMOD::CreateSoundRenderer |                            | FMOD::System                |     |

```
//
FMOD::System* pFMOD = NULL;
result = FMOD::System_Create(&pFMOD);
if (result != FMOD_OK)
    exit(1);
result = pFMOD->getVersion(&version);
if (result != FMOD_OK || version < FMOD_VERSION)
    exit(1);
result = pFMOD->init(64, FMOD_INIT_NORMAL, 0);
if (result != FMOD_OK)
    exit(1);
GPtr<GSoundRenderer> psoundRenderer =
    *GSoundRendererFMOD::CreateSoundRenderer();
if (!psoundRenderer->Initialize(pFMOD))
    exit(1);

//      Gfx::Audio
Ptr<Gfx::Audio> paudioState = *new Gfx::Audio(psoundRenderer);
loader.SetAudio(paudioState);
```

## 8.2

Scaleform

[Video](#)

Gfx::Loader

[VideoSoundSystem](#)

Gfx::Video::Video

Scaleform

SoundRenderer

```
//          Video
Ptr<Gfx::Video::Video> pvc = *new Video(Thread::NormalPriority);
//          SoundRenderer
if (psoundRenderer)
{
    pvc->SetSoundSystem(psoundRenderer);
}
//
//          SetSoundSystem          Video
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new
                                                    VideoSoundSystemDX8(0)));

loader.SetVideo(pvc);
```

Windows ,Video::Video VideoPC

affinity mask

```
VideoPC(Thread::ThreadPriority
        decodingThreadsPriority = Thread::NormalPriority,
        int decodingThreadsNumber = MAX_VIDEO_DECODING_THREADS,
        UInt32* affinityMask = NULL);
```

```
UInt32 affinities[] = {
    DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
    DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
    DEFAULT_VIDEO_DECODING_AFFINITY_MASK };
```

```
Ptr<Video::Video> pvc = *new Video::VideoPC(Thread::NormalPriority,
                                              MAX_VIDEO_DECODING_THREADS, affinities);
```

PS3 Xbox360

Scaleform

Video

```
VideoPS3(CellSpurs* spurs,
          int ppu_num, Thread::ThreadPriority ppu_priority,
          int spu_num, UInt8* spurs_priorities = NULL);
```

```
spurs -          CellSpurs
spu_num -          SPU
ppu_num -          PPU
ppu_priority -    PPU
spurs_priorities - SPU
```

```
UInt8 spurs_priorities[] = {1,1,1,1,0,0,0,0};
Ptr<Video::Video> pvc = *new VideoPS3(GetSpurs(), 0, 1000,
                                     4, spurs_priorities);
```

### PPU

```
Scaleform Video          PPU          ,          50-60 FPS          720p          ,PPU
5%          ,
,
,PPU          ,
40%    PPU          ,          ,          ,
4-5%          ,          ,          1%
```

```
UInt8 spurs_priors[] = {0,0,1,1,1,1,0,0};
Ptr<Video::Video> pVideo = *new VideoPS3(GetSpurs(), 0,
                                     Thread::BelowNormalPriority, 4, spurs_priors);
```

```
Xbox360          ,          VideoXbox360,
,          ,
```

```
VideoXbox360(unsigned decodingProcs = Xbox360_Proc1 | Xbox360_Proc3 |
          Xbox360_Proc5, Thread::ThreadPriority
          decodingThreadsPriority = Thread::NormalPriority,
          Xbox360Proc readerProc = Xbox360_Proc2);
```

```
Ptr<Video::Video> pvc = *new VideoXbox360(Xbox360_Proc1 |
          Xbox360_Proc3|Xbox360_Proc5,
          Thread::NormalPriority,
          Xbox360_Proc2);
```

## 8.3

## API

Video::VideoBase::ReadCallback –

VideoBase::IsIORequired VideoBase::EnableIO

*ReadCallback*

```
//
class VideoReadCallback : public VideoBase::ReadCallback
{
    public:
    VideoReadCallback() {}
    virtual ~VideoReadCallback() {}

    virtual void OnReadRequested()
    {
        //
        VideoReadStart = Timer::GetTicks();
        // ...
    }
    virtual void OnReadCompleted()
    {
        //
        GameReadStart = Timer::GetTicks();
        // ...
    }

    UInt64 VideoReadStart;
    UInt64 GameReadStart;
};

//
Ptr<Video::Video> pvc = *new Video::VideoPC(
    Thread::NormalPriority, MAX_VIDEO_DECODING_THREADS, affnts);

//
Ptr<VideoReadCallback> pvideoReadCallback = *new VideoReadCallback;
pvc->SetReadCallback(pvideoReadCallback);
```

*IsIORequired*

*EnableIO*

*EnableIO*

/

```
if(pvc->IsIORequired())
{
    pDataLoader->Enable(false);
    pvc->EnableIO(true);
    fprintf(stderr, "Video is loading\n");
}

// ...

if(!pvc->IsIORequired())
{
    pvc->EnableIO(false);
    pDataLoader->Enable(true);
    fprintf(stderr, "Game data is loading\n");
}
```

VideoBGLoadFlash    VideoBGLoadData

*Apps\Samples\Video*

- VideoBGLoadFlash – Flash
- VideoBGLoadData –

## 8.4 Scaleform VideoSoundSystem

[GFX::Video::VideoSoundSystem](#)

Video::Video playback

[Video::SetSoundSystem\(\)](#)。

Video :

- Video::VideoSoundSystemDX8 – Windows 中的 DirectSound
- Video::VideoSoundSystemXA2 – Windows 與 Xbox360 中的 XAudio2

- Video::VideoSoundSystemPS3 – MultiStream for PS3
- Video::VideoSoundSystemWii – Wii 系統音頻
- Video::VideoSoundSystemFMOD - 基於 FMOD 的音頻介面
- Video::VideoSoundSystemWwise – 基於 Wwise 的音頻介面

Scaleform

Flash

Video::VideoSoundSystem

Scaleform

Sound::SoundRenderer

8.1

VideoSoundSystem

VideoSound

SoundRenderer

SoundRenderer

Video

```
#include "Video/Video_VideoSoundSystemXA2.h"
```

```
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));
```

[Gfx::Video::VideoSoundSystem](#)

[Gfx::Video::VideoSound](#)

VideoSoundSystem

VideoSoundSystem

Create

VideoSound

Scaleform

VideoSound

Scaleform

VideoSound::PCMStream

VideoSoundSystem

VideoSoundSystemWwise

Wwise SDK

v2009.2.1 build 3271. Audiokinetic

Visual Studio /

SDK\samples\Plugins\AkAudioInput.

Wwise

Gfx::Video::Video

Wwise

SDK

```
#include "Video/Video_VideoSoundSystemWwise.h"
```

```
Ptr<Video::VideoSoundSystem> wwise =
```

```
Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemWwise());
```

```
pvc->SetSoundSystem(wwise);
```

```
wwise->Update();
```

Scaleform Player - FxPlayerAppBase.cpp

FxSoundFMOD.cpp

FxSoundWwise.cpp

Apps\Samples\FxPlayer

Apps\Samples\Common

Video\_VideoSoundSystem\*.cpp

Src\Video

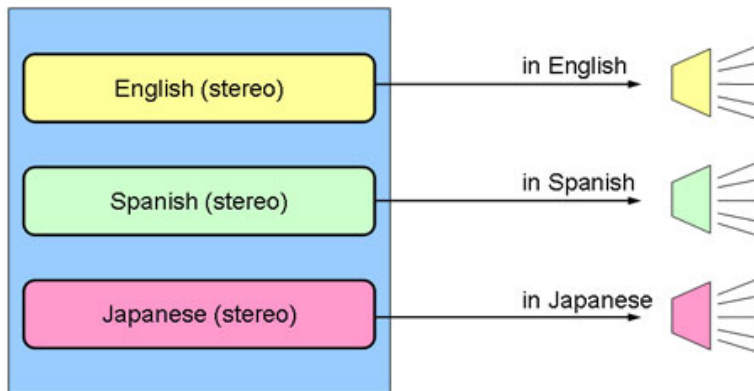
## 8.5

```

Gfx::Video::Video
ActionScript
,
NetStream
audioTrack
,
: ns.audioTrack = 2;

```

Multilingual Movie Data



12

5.1

Gfx::Video::Video

- 1.
2. SubAudio

### 8.5.1

```

,
5.1
,
5.1
,
5.1
,
5.1

```

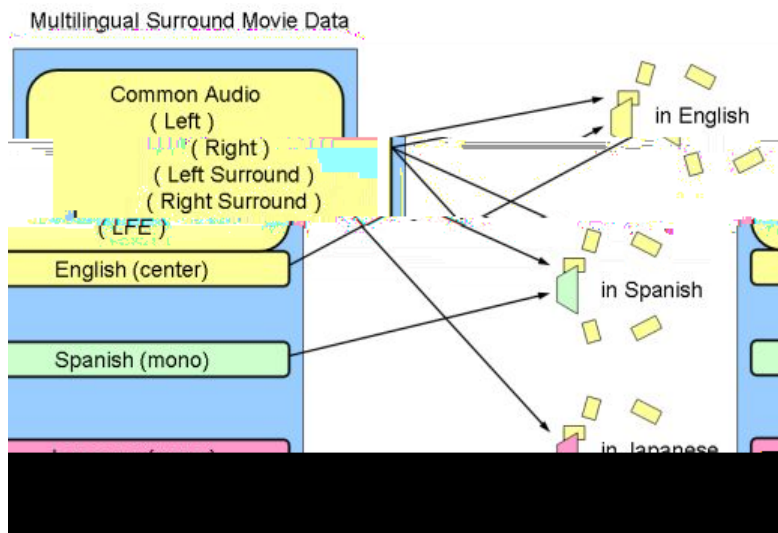
```

ns.audioTrack = 0; // main 5.1 audio track
ns.voiceTrack = 5; // mono audio track

```



5.1



13

## 8.5.2 SubAudio

SubAudio , GFX::Video::Video  
 ,GFX::Video::Video  
 SubAudio

```
ns.audioTrack = 0; // main audio track
ns.subAudioTrack = 6; // secondary audio track
```

: SubAudio Playback

Sound.setVolume()

```
var audio:Sound = new Sound(this);
audio.setVolume(80, 50); // set volume for main and subaudio tracks
```

## 14 SubAudio

## 9

, (Stuttering),  
 , :

1. ( IO ),

2. ,

3. ( )

, ( ),

1. - , ,  
 (720P ) 3 4 ,  
 8

2. - /  
 Scaleform

3. - ActionScript , 6  
 a. `NetStream.setNumberOfFramePools(numPools:Number)`  
 b. `NetStream.setBufferTime(bufferTime:Number)`